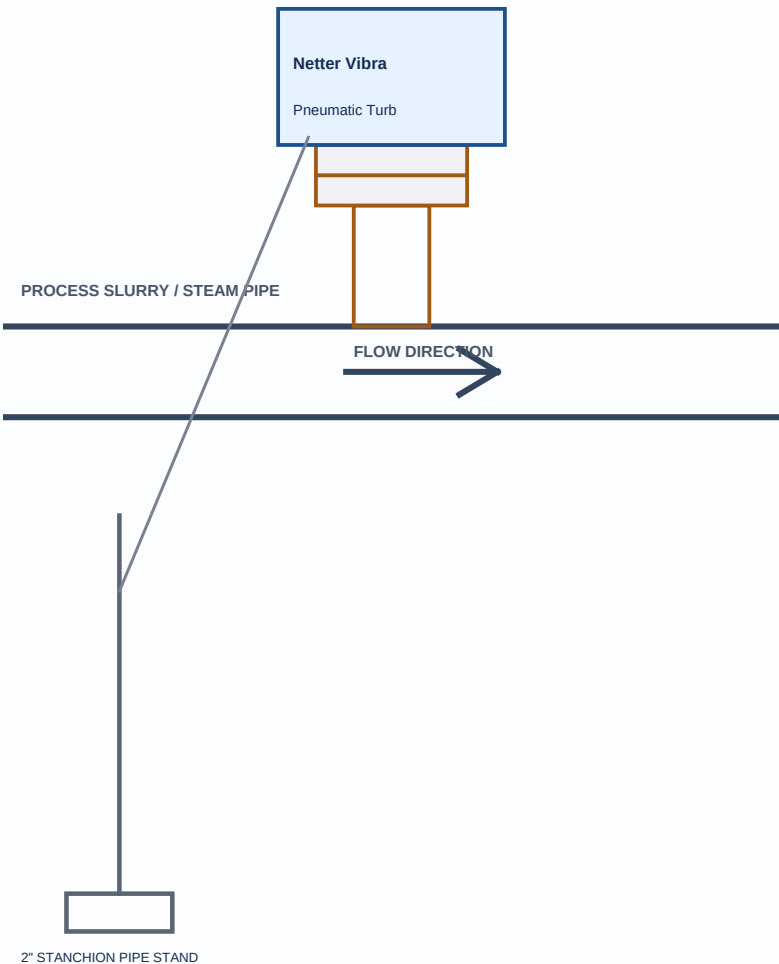


TEMPLATE TYP-23: PNEUMATIC TURBINE VIBRATOR

TYPICAL TAGS: XV-60901 Series

SERVICE: Powder Silo & Hopper Wall Discharge Agitation | PLC I/O: 1756-OB32 (24V DC Digital Output)

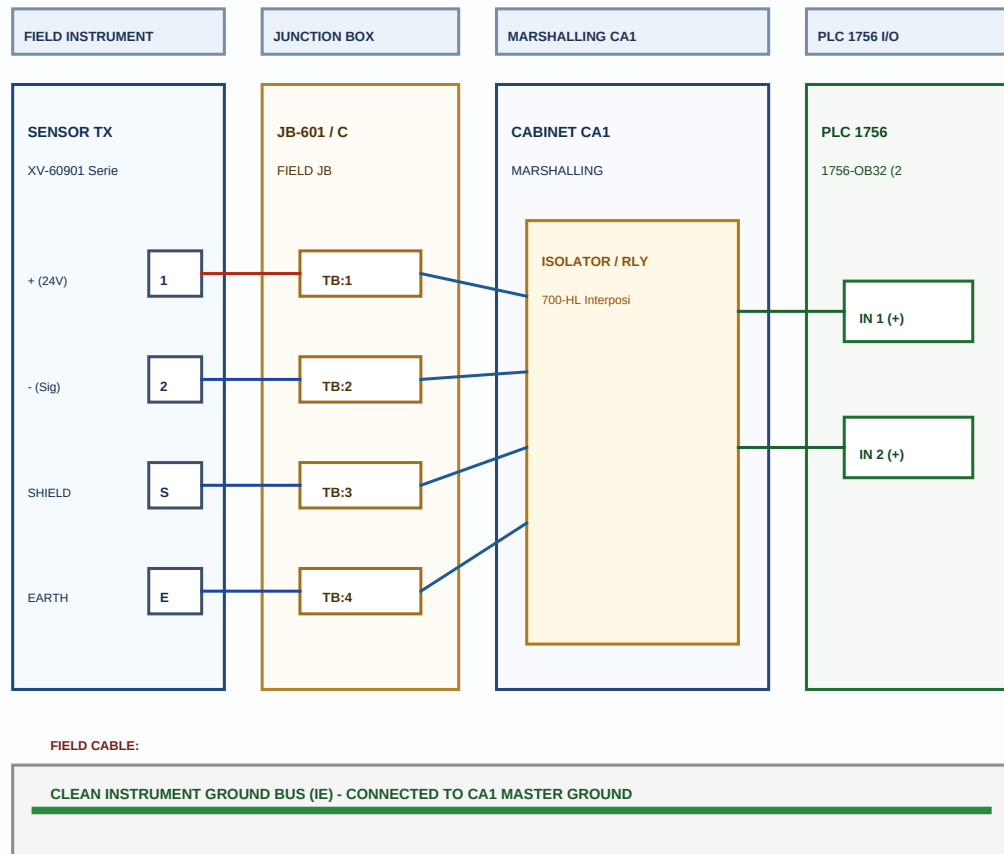
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	G 1/8" BSP Air Port / Rigid ..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	Extruded Aluminum Hard Anodi..
Hazardous Class:	ATEX II 2GD c T6 85°C (Mecha..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	DO: Pulsed Vibration Burst (Timer Cont..
Power Supply:	24V DC from 1756-OB32 + 2 to 6 bar Lub..
Field Cable Type:	3Cx1.5mm² Shielded Cable (Belden 9533)
Destination JB:	JB-601 / CA1 Marshalling
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	700-HL Interposing Relay with Diode
PLC Module:	1756-OB32 (24V DC Digital Output)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: NETTER VIBRATION

MODEL: NCT 5 Pneumatic Turbine Vibrator

SERVICE: Powder Silo & Hopper Wall Discharge Agitation

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Pneumatic Turbine Vibrator NCT Netter Vibration High-Speed Unit		1 EA
2	Solenoid Valve 3/2-Way 24VDC (High Flow Fast-Cycling Solenoid		1 EA
3	Silencer / Exhaust Air Filter Sintered Bronze Sinter Filter		1 EA
4	SMC Air Filter Regulator Kit Modular AFR w/ Pressure Gauge		1 SET
5	Weld-on Hopper Mounting Channe Reinforced Channel Profile Steel		1 EA

ENGINEERING INSTALLATION NOTES

- Mount vibrator on stiffened channel beam welded to hopper cone to transmit vibration into bulk powder.
- Operate in intermittent pulse mode (e.g. 5 seconds ON, 30 seconds OFF) via PLC timer logic.
- Interposing relay 700-HL at CA1 panel protects PLC output module from inductive flyback.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTPNEUMATIC TURBINE VIBRATOR HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.AEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONDWG NO: KAL-JC-TYP-023
REV: 3.6 DATE: 2026-09-08PROJECT NO:
R5THS00138-JCSHEET NO:
25 / 25